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Elektronika Analog

LATIHAN SOAL BAB 3 No. 1 – 39

1. What names are applied to the two types of BJT transistors? Sketch the basic construction of each and label the various minority and majority carriers in each. Draw the graphic symbol next to each. Is any of this information altered by changing from a silicon to a germanium base?

Jawaban

Kedua jenis Bipolar Junction Transistor (BITS) adalah,

•Transistor PNP

•Transistor NPN Pada transistor PNP, semikonduktor tipe-n ditempatkan di antara dua semikonduktor tipe-p pada transistor NPN, semikonduktor tipe-p ditempatkan di antara dua semikonduktor tipe n

2. What is the major difference between a bipolar and a unipolar device?

Jawaban

Perbedaan utama antara transistor bipolar dan unbipolar ialah transistor bipolar memanfaatkan elektron dan lubang dalam proses injeksi atau proses muatan yang terdiri dari dua jenis transistor serta perubahan listriknya dalam jumlah kecil dapat menghasilkan perubahan arus listrik jumlah besar pada terminal kolektor. Sedangkan transistor unbipolar hanya memanfaatkan salah satunya dari elektron atau lubang dalam proses muatan.

3. How must the two transistor junctions be biased for proper transistor amplifier operation?

Jawaban

Bias mundur dan bias maju

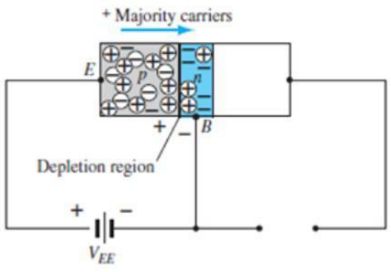
4. What is the source of the leakage current in a transistor?

Jawaban

Sumber arus bocor (Io) pada transistor ialah arus pembawa minoritas di kolektor

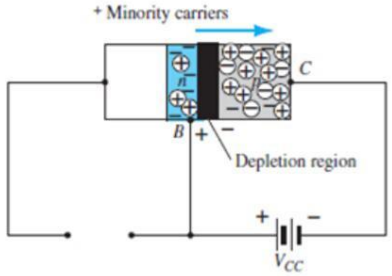
5. Sketch a figure similar to Fig. 3.3 for the forward-biased junction of an npn transistor. Describe the resulting carrier motion.

Jawaban:



6. Sketch a figure similar to Fig. 3.4 for the reverse-biased junction of an npn transistor. Describe the resulting carrier motion.

Jawaban:



7. Sketch a figure similar to Fig. 3.5 for the majority- and minority-carrier flow of an npn transistor. Describe the resulting carrier motion.
Jawaban

8. Which of the transistor currents is always the largest? Which is always the smallest? Which two currents are relatively close in magnitude?
Jawaban
I_E terbesar
I_B terkecil
$I_C \cong I_E$

9. If the emitter current of a transistor is 8 mA and I_B is 1/100 of I_C , determine the levels of I_C and I_B.
Jawaban
$I_B = \frac{1}{100} Ic \rightarrow Ic = 100 I_B$
$I_E = I_C + I_B = 100 I_B + I_B = 101 I_B$
$I_B = \frac{IE}{101} = \frac{8\,mA}{101} = 7.921 \mu A$

$I_C = 100 I_B = 100\ (79{,}21\ \mu A) = 7{,}921\ mA$

10. From memory, sketch the transistor symbol for a pnp and an npn transistor, and then insert the conventional flow direction for each current
Jawaban

11. Using the characteristics of Fig. 3.7 , determine V_{BE} at I_E 5 mA for V_{CB} = 1,10, and 20 V. Is it reasonable to assume on an approximate basis that V_{CB} has only a slight effect on the relationship between V_{BE} and I_E ?
Jawaban
$I_E = 5mA, V_{CB} = 1V\ ;\ V_{BE} = 800mV$
$V_{CB} = 10\ V\ ; V_{BE} = 770\ mV$
$V_{CB} = 20\ V\ ;\ V_{BE} = 750\ mV$
Perubahan dari Vcb ialah 20 volt : 1 volt = 20 : 1

12. a. Determine the average ac resistance for the characteristics of Fig. 3.10b .
b. For networks in which the magnitude of the resistive elements is typically in kilohms, is the approximation of Fig. 3.10c a valid one [based on the results of part (a)]?
Jawaban
a. $r_{av} = \frac{\Delta V}{\Delta I} = \frac{0{,}9\ V - 0{,}7\ V}{8\ mA - 0} = 25\ \Omega$

b. Ya, karena nilai resistansi dari 25 Ω diabaikan dibanding dengan nilai resistansi lainnya

13. a. Using the characteristics of Fig. 3.8 , determine the resulting collector current if I_E = 4.5 mA and V_{CB} = 4 V.

b. Repeat part (a) for I_E 4.5 mA and V_{CB} 16 V.

c. How have the changes in V_{CB} affected the resulting level of I_C ?

d. On an approximate basis, how are I_E and I_C related based on the results above?

Jawaban

a. $I_C \cong I_E = 4{,}5\ mA$

b. $I_C \cong I_E = 4{,}5\ mA$

c. Perubahan Vcb yang mempengaruhi tingkat Ic, diabaikan karena perubahan tersebut tidak dapat dideteksi pada rangkaian ini

d. $I_C \cong I_E$

14. a. Using the characteristics of Figs. 3.7 and 3.8 , determine I_C if V_{CB} = 10 V and V_{BE} = 800mV.

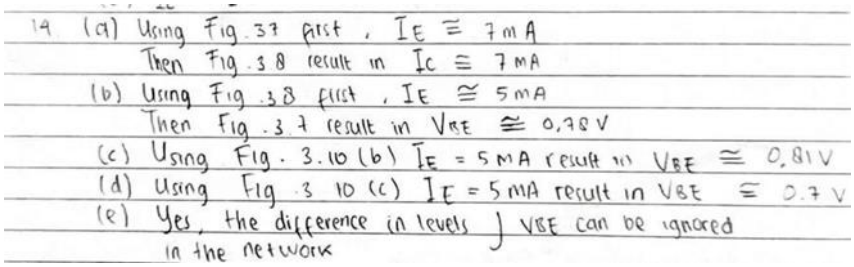
b. Determine V_{BE} if I_C = 5 mA and V_{CB} = 10 V.

c. Repeat part (b) using the characteristics of Fig. 3.10b .

d. Repeat part (b) using the characteristics of Fig. 3.10c .

e. Compare the solutions for V_{BE} for parts (b) through (d). Can the difference be ignored if voltage levels greater than a few volts are typically encountered?

Jawaban



15. a. Given an α_{dc} of 0.998, determine I_C if I_E = 4 mA.

b. Determine α_{dc} if I_E = 2,8 mA and I_B = 20 µA

c. Find I_E if I_B = 40 µA and is 0,98.

Jawaban

a. $I_C = \alpha\ I_E$= (0.998)(4 mA) = 3.992 Ma

b.

16. From memory only, sketch the common-base BJT transistor configuration (for npn and pnp) and indicate the polarity of the applied bias and resulting current directions.

Jawaban

17. Calculate the voltage gain ($A_v = V_L/V_i$) for the network of Fig. 3.12 if Vi = 500 mV and R = 1 kΩ. (The other circuit values remain the same.)

Jawaban

Ii = $\frac{V1}{R1} = 500\frac{MV}{20\Omega}$ = 25 mA

I2 = I1=25 mA

V2 = I1 R1

= (25 Ma)(1KΩ)

= 25 V

Ao = $\frac{Vo}{V1} = \frac{25\,v}{0{,}5v}$ = 50

18. Calculate the voltage gain ($A_v = V_L/V_i$) for the network of Fig. 3.12 if the source has an internal resistance of 100 Ω in series with Vi.

Jawaban

Ii = $\frac{Vi}{R1+R3} = \frac{200\,mV}{20\Omega+100\Omega} = \frac{200\,mV}{120\Omega}$ = 1,67 mA

I2 = Ii =1,67 mA

V2 = I.R = (1,67 mA)(5KΩ) = 35 V

Av = $\frac{Vo}{Io} = \frac{0{,}35\,V}{O{,}2\,V}$ = 41,75

19. Define I_{CBO} and I_{CEO} . How are they different? How are they related? Are they typically close in magnitude?

20. Using the characteristics of Fig 3.14:

(a) Find the value of *IC* corresponding to *VBE* = +750 mV and *VCE* = +5 V

→ *IB* = 35μA, *IC* = 5,2 mA

(b) Find the value of *VCE* and *VBE* corresponding to *IC* = 3 mA and *IB* = 30 μA.

→ *VCE* = 13 V, *VBE* = 0.75 V

21. (a) For the common-emitter characteristics of Fig. 3.14, find the dc beta at an operating point

of *VCE* = +8 V and *IC* = 2 mA

→ ßdc = $\frac{Ic}{IB} = \frac{2\,mA}{18\mu A}$ = 111.11

(b) Find the value of α corresponding to this operating point.

→ α = $\frac{ß}{ß+1} = \frac{111.11}{111.11+1}$ = 0.991

(c) Calculate the approximate value of *ICBO* using the dc beta value obtained in part (a).

→ *ICBO* = (1 – α) *ICEO* = (1 – 0.991) (0.3 mA) = 27 μA

22. (a) Using the characteristics of Fig. 3.14a, determine *ICEO* at *VCE* = 10 V.

→ *ICEO* = 0.3 mA

(b) Determine ßdc at *IB* = 10 μA and *VCE* = 10 V

→ *IC* = 1.35 mA

ßdc = $\frac{Ic}{IB} = \frac{1.35\,mA}{10\,\mu A}$ = 135

(c) Using the ßdc determined in part (b), calculate I_{CBO}.

→ $\alpha = \frac{ß}{ß+1} = \frac{135}{136} = 0.9926$

$I_{CBO} = (1-\alpha)\, I_{CEO}$

$= (1-0.9926)(0.3mA) = 2.2\ \mu A$

23. (a) Using the characteristics of Fig. 3.14a, determine ßdc at $I_B = 80\ \mu A$ and $V_{CE} = 5$ V

→ $ßdc = \frac{Ic}{IB} = \frac{6.85\ mA}{80\ \mu A} = 85{,}6$

(b) Repeat part (a) at $I_B = 5\ \mu A$ and $V_{CE} = 15$ V

→ $ßdc = \frac{Ic}{IB} = \frac{1\ mA}{5\ \mu A} = 2$

(c) Repeat part (a) at $I_B = 30\ \mu A$ and $V_{CE} = 10$ V

→ $ßdc = \frac{Ic}{IB} = \frac{3.35\ mA}{30\ \mu A} = 111{,}6$

(d) Reviewing the results of parts (a) through (c), does the value of ßdc change from point to point on the characteristics? Where were the higher values found? Can you develop any general conclusions about the value of ßdc on a set of characteristics such as those provided in Fig. 3.14a

→ level tertinggi pada nilai Ic

24. (a) Using the characteristics of Fig. 3.14a, determine ßac at $I_B = 80\ \mu A$ and $V_{CE} = 5$ V

→ $ßac = \frac{\Delta Ic}{\Delta IB} = \frac{7.5\ mA - 6.2\ mA}{90\ \mu A - 70\ \mu A} = 65$

(b) Repeat part (a) at $I_B = 5\ \mu A$ and $V_{CE} = 15$ V

→ $ßac = \frac{\Delta Ic}{\Delta IB} = \frac{1.2\ mA - 0.4\ mA}{10\ \mu A - 0\ \mu A} = 80$

(c) Repat part (a) at $I_B = 30\ \mu A$ and $V_{CE} = 10$ V

→ $ßac = \frac{\Delta Ic}{\Delta IB} = \frac{4.2\ mA - 2.3\ mA}{40\ \mu A - 20\ \mu A} = 95$

(d) Reviewing the results of parts (a) through (c), does the value of ßac change from point to point on the characteristics? Where are the high values located? Can you develop any general conclusions about the value of ßac on a set of collector characteristic ?

→ nilai tertinggi terdapat pada Ic

(e) The chosen points in this exercise are the same as those employed in Problem 23. If Problem 23 was performed, compare the levels of ßdc and ßac for each point and comment on the trend in magnitude for each quantity

→ ßdc > ßac

25. Using the characteristics of Fig. 3.14a, determine ßdc at I_B = 25 µA and V_{CE} = 10 V. Then calculate αdc and the resulting level of I_E. (Use the level of I_C determined by I_C = ßdc I_B)

→ ßdc $= \frac{Ic}{IB} = \frac{2.9\ mA}{25\ \mu A} = 116$

α $= \frac{ß}{ß+1} = \frac{116}{116+1} = 0.991$

$I_E = \frac{Ic}{\alpha} = \frac{2.9\ mA}{0.991} = 2.93$ mA

26. (a) Given that αdc = 0.987, determine the corresponding value of ßdc

→ ß $= \frac{\alpha}{1-\alpha} = \frac{0.987}{1-0.987} = 75{,}92$

(b) Given ßdc = 120, determine the corresponding value of α

→ α $= \frac{ß}{ß+1} = \frac{120}{120+1} = 0.992$

(c) Given that ßdc = 180 and I_C = 2.0 mA, find I I_E and I_B.

→ $I_B = \frac{Ic}{ß} = \frac{2\ mA}{180} = 11.11\mu A$

$I_E = I_C + I_B$ = 2 mA + 11.11 µA = 2.011 mA

27. From memory, and memory only, sketch the common-emitter configuration (for npn and pnp) and insert the proper biasing arrangement with the resulting current directions for I_B, $I_C + I_E$.

28. 2An input voltage of 2 V rms (measured from base to ground) is applied to the circuit of Fig.3.21. Assuming that the emitter voltage follows the base voltage exactly and that Vbe (rms) = 0.1 V, calculate the circuit voltage amplification (Av = Vo /Vi) and emitter current for RE = 1 kΩ.

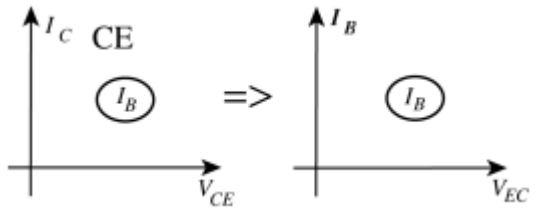
→ Ve = Vi – Vbe = 2 V – 0.1 V = 1.9 V

Av = $\frac{Vo}{Vi} = \frac{1.9\ V}{2\ V} = 0.95 = 1$

Ie = $\frac{Ve}{Re} = \frac{1.9\ V}{1\ K\Omega} = 1.9$ mA

29. For a transistor having the characteristics of Fig. 3.14, sketch the input and output characteristics of the common-collector configuration.

→ karakteristik output :



Karakteristik input :

Common-emitter input dapat digunakan secara lansung untuk perhitungan common-collector

30. Determine the region of operation for a transistor having the characteristics of Fig. 3.14 if

ICmax = 7 mA, *VCEmax* = 17 V, and *PCmax* = 40 mW

→ *PCmax* = 40 mW = *VCEIC*

IC = *ICmax*, *VCE* = $\frac{PCmax}{ICmax} = \frac{40\ mW}{7\ mA} = 5.71$ V

VCE = *VCEmax*, *IC* = $\frac{PCmax}{VCEmax} = \frac{40\ mW}{17\ V} = 2.35$ mA

VCE = V, *IC* = $\frac{PCmax}{VCE} = \frac{40\ mW}{V} =$ mA

IC = mA, *VCE* = $\frac{PCmax}{IC} = \frac{40\ mW}{mA} =$ V

VCE = V, *IC* = $\frac{PCmax}{VCE} = \frac{40\ mW}{V} =$ mA

31. Determine the region of operation for a transistor having the characteristics of Fig. 3.8 if IC_{max} = 6 mA, V_{CB}max = 15 V, and PC_{max} = 30 mW.

→ *IC* = *ICmax*, *VCE* = $\frac{PCmax}{ICmax} = \frac{30\ mW}{6\ mA} = 5$ V

VCB = *VCBmax*, *IC* = $\frac{PCmax}{VCBmax} = \frac{30\ mW}{15\ V} =$ mA

IC = , *VCE* = $\frac{PCmax}{IC} = \frac{40\ mW}{mA} =$ V

32. Referring to Fig. 3.23, determine the temperature range for the device in degrees Fahrenheit

→ temperature akan beroperasi dalam -55°C≤ Tj ≤ 150°C

°F = $\frac{9}{5}$°C + 32°

$= \frac{9}{5}(150^{\circ}C) + 32^{\circ} = -67^{\circ}F$

$^{\circ}F = \frac{9}{5}(150^{\circ}C) + 32^{\circ} = 302^{\circ}F$. Jadi, $-67^{\circ}C \leq Tj \leq 302^{\circ}C$

33. Using the information provided in Fig. 3.23 regarding *PDmax, VCEmax, ICmax and VCEsat*, sketch the boundaries of operation for the device

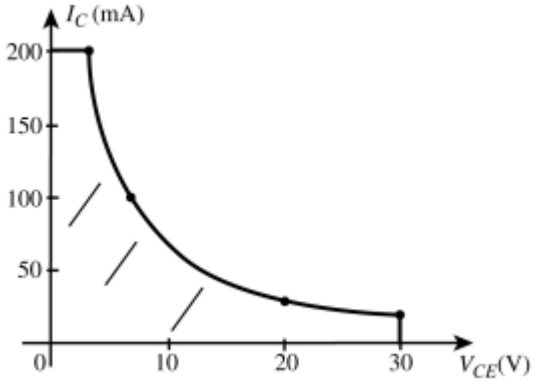
→ *ICmax* = 200 mA ,*VCEmax* = 30 V, *PDmax* = 625 mW

$IC = ICmax, VCE = \frac{PDmax}{ICmax} = \frac{625\ mW}{200\ mA} = 3.125$ V

$VCE = VCEmax, IC = \frac{PDmax}{VCEmax} = \frac{625\ mW}{30\ V} =$ 20.83 Ma

$IC = 100$ mA , $VCE = \frac{PDmax}{IC} = \frac{625\ mW}{100\ mA} =$ 6.25 V

$VCE = 20$ V , $IC = \frac{PDmax}{VCE} = \frac{635\ mW}{20\ V} =$ 31.25 mA



34. Based on the data of Fig. 3.23, what is the expected value of ICEO using the average value of ßdc?

→ *ICBO* = 50 nA max

$ßavg = \frac{ß\min + ßmax}{2} = \frac{50+150}{2} = \frac{200}{2} = 100.$ Jadi, *ICEO = ßicbo* = (100) (50 nA) = 5µA

35. How does the range of *hFE* (Fig. 3.23(j), normalized from *hFE* = 100) compare with the range of *hFE* (Fig. 3.23(f)) for the range of IC from 0.1 to 10 mA?

→

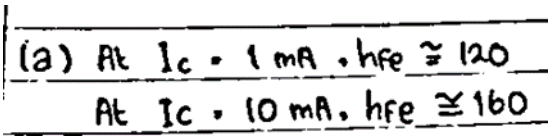
36. Using the characteristics of Fig. 3.23b, determine whether the input capacitance in the common-base configuration increases or decreases with increasing levels of reverse-bias potential. Can you explain why?

Jawaban

Ketika potensi bias balik meningkat besarnya, kapasitansi input Cibo berkurang (Gbr. 1). 3.23(b)). Peningkatan potensi bias balik menyebabkan lebar daerah penipisan menjadi meningkat, sehingga mengurangi kapasitansi ($C = \varepsilon\frac{A}{d}$) .

37. Using the characteristics of Fig. 3.23f, determine how much the level of hfe has changed from

its value at 1 mA to its value at 10 mA. Note that the vertical scale is a log scale that may require reference to Section 11.2. Is the change one that should be considered in a design situation?
Jawaban
(b) Hasil tersebut mengkonfirmasi kesimpulan dari soal 21 dan 22 bahwa beta cenderung meningkat dengan meningkatnya arus kolektor.
38.Using the characteristics of Fig. 3.23j, determine the level of βdc at IC = 10 mA at the three levels of temperature appearing in the figure. Is the change significant for the specified temperature range? Is it an element to be concerned about in the design process?



39. (a)Using the characteristics of Fig. 3.24, determine ac at IC = 14 mA and VCE = 3 V.

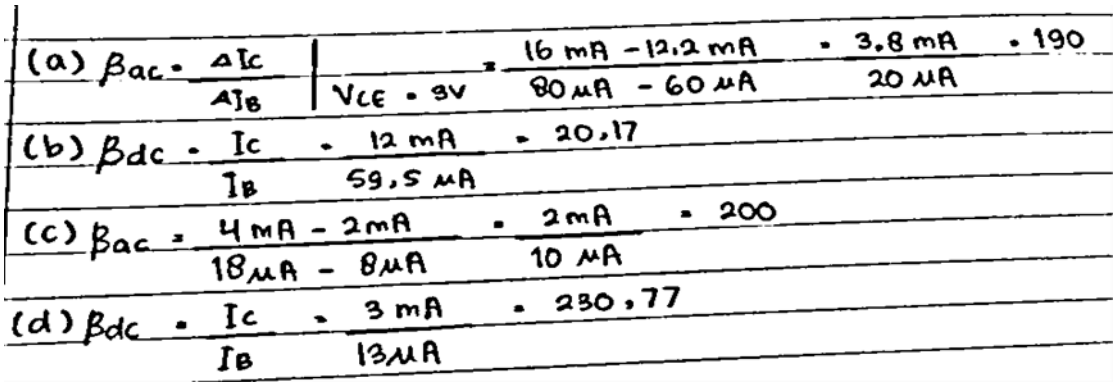
(b) Determine βdc at IC = 1 mA and VCE = 8 V.

(c) Determine βac at IC = 14 mA and VCE = 3 V.

(d) Determine βdc at IC = 1 mA and VCE = 8 V.

(e) How does the level of βac and βdc compare in each region?

(f) Is the approximation βdc ≅ βac a valid one for this set of characteristics?



(e) Dalam kedua kasus βdc sedikit lebih tinggi dari βac (≅10%)

(f) (g)

Secara umum βdc danβ ac meningkat dengan meningkatnya IC untuk VCE tetap dan keduanya menurun untuk penurunan tingkat VCE untuk IE tetap. Namun, jika IC meningkat sementara VCE menurun ketika bergerak di antara dua titik pada karakteristik, peluangnya adalah tingkat βdc atau βac mungkin tidak berubah secara signifikan. Dengan kata lain, peningkatan yang diharapkan karena peningkatan arus kolektor dapat diimbangi dengan penurunan VCE. Data di atas mengungkapkan bahwa ini adalah kemungkinan yang kuat karena level β relatif dekat